SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:	Amirtha Sri N	**Reg. No.:**	1911260175
Date:	27.10.2026	**Duration:**	60 minutes

Code of Conduct

I AmirthaSri (1911260175)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:
Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**

- Identify all primary and secondary actors interacting with the system.
- List the major use cases associated with each actor.
- Draw a Use Case Diagram representing the interactions between actors and the system.

4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description
   - Scope of the System
   - Functional Requirements
   - Non-Functional Requirements
   - Use Case Descriptions
   - Data Requirements
   - External Interface Requirements
   - System Features
5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.

Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

AT1 – SCENARIO-BASED REQUIREMENT ANALYSIS

Project: AI-Based College FAQ and Student Support Chatbot

1. Introduction

The AI-Based College FAQ and Student Support Chatbot is a software system designed to provide quick and reliable answers to common questions asked by college students. Students often need information about admissions, departments, class schedules, examinations, fees, events, attendance, facilities, and college rules. At present, they may have to contact different offices or wait for staff members to respond. The proposed chatbot provides a single digital platform where students can ask questions in simple language and receive relevant answers.

The purpose of this Software Requirement Specification (SRS) is to describe the system's objectives, scope, users, functional requirements, non-functional requirements, interfaces, assumptions, and constraints.

2. Scenario Analysis

The system belongs to the education and student-support domain. A college receives many repeated questions from students, parents, and applicants. Staff members spend time answering routine queries, and students may experience delays or receive information from unofficial sources. The chatbot will use a maintained set of college information and AI-based language processing to understand user questions and return suitable responses. If a question is unclear or cannot be answered confidently, the system will direct the user to the appropriate department or support staff.

Main stakeholders are current students, prospective students, parents or guardians, college administrative staff, department coordinators, and the system administrator. Their common objective is to access accurate information quickly and reduce the workload involved in answering repeated questions.

3. Problem Description

Students may find it difficult to locate correct college information because details are spread across notices, websites, office counters, and messages. Administrative staff may receive the same questions repeatedly, especially during admission, examination, and fee-payment periods. Manual responses can be delayed outside office hours. Therefore, a centralized chatbot is required to answer frequently asked questions, guide students to relevant information, and provide a clear route to human support when necessary.

4. Objectives of the System

- Provide quick answers to frequently asked college-related questions.

- Make essential student-support information available through one interface.
- Reduce repetitive enquiries handled by administrative staff.
- Help users find the correct department or official contact for unresolved questions.
- Keep responses consistent by using verified and regularly updated college information.
- Provide a simple interface that can be used by students with different levels of technical knowledge.

5. Scope of the System

The system will answer questions related to college facilities, departments, admissions, academic calendars, class and examination schedules, fees, events, attendance rules, and general student services, provided that the relevant information has been entered and verified. It will allow users to type questions, view answers, and access department contact details or support instructions. Authorized administrators will be able to add, edit, review, and remove FAQ content.

The chatbot will not independently approve admissions, change marks, make fee transactions, alter attendance records, or make official academic decisions. Such requests must be handled through the college's authorized systems and staff.

6. Stakeholders

- Students: ask questions and receive college-related information.
- Prospective students: seek information about courses, admissions, and facilities.
- Parents or guardians: obtain general information and official contact details.
- College administrative staff: provide verified information and handle queries requiring human assistance.
- Department coordinators: review department-specific details and help keep information current.
- System administrator: manage users, FAQ records, access permissions, and system maintenance.

7. Functional Requirements

Functional requirements describe the actions that the system must perform.

- FR1 – User Question Input: The system shall allow users to enter a question through a chat interface.
- FR2 – Question Processing: The system shall analyze the question and identify its likely topic or intent.
- FR3 – Answer Generation: The system shall provide an answer based on approved college information and available FAQ records.
- FR4 – Suggested Questions: The system shall display common questions or suggestions to help users begin a conversation.
- FR5 – Clarification: If a question is unclear, the system shall ask the user to provide more details.
- FR6 – Unanswered Queries: If the system cannot provide a reliable answer, it shall explain this and direct the user to the relevant office or support contact.

- FR7 – FAQ Management: Authorized administrators shall be able to add, edit, review, and delete FAQ entries.
- FR8 – Information Updates: The system shall allow approved college information to be updated when schedules, rules, contacts, or notices change.
- FR9 – Conversation Handling: The system shall display the current conversation clearly and allow the user to ask follow-up questions.
- FR10 – Access Control: The system shall restrict FAQ-management functions to authorized administrators.
- FR11 – Feedback: The system may allow users to indicate whether an answer was helpful.
- FR12 – Query Records: The system may maintain limited query records for improving FAQ coverage, subject to privacy rules.

8. Non-Functional Requirements

- Performance: The system should respond to ordinary questions within a few seconds under normal operating conditions.
- Usability: The interface should be simple, readable, consistent, and easy for first-time users.
- Reliability: The system should provide dependable responses based on verified information and clearly identify queries it cannot answer.
- Security: Administrative functions must require authorized access, and sensitive information must be protected.
- Privacy: The system should collect only the information needed for its operation and should not request unnecessary personal details.
- Availability: The chatbot should be available whenever the hosting service and network are operational, including outside office hours.
- Scalability: The system should support an increase in users and FAQ records without major redesign.
- Maintainability: FAQ content, contact details, and system settings should be easy for authorized staff to update.
- Compatibility: The interface should work on commonly used web browsers and mobile devices.
- Accuracy: Answers should be based on current, approved college information; uncertain answers should be referred to human support.

9. Actors and Use Cases

Primary actors:

- Student / Prospective Student – asks questions, views answers, uses suggested questions, and requests support information.
- Parent or Guardian – asks general questions about admissions, facilities, schedules, and official contacts.
- Administrator – signs in and manages FAQ entries and approved information.
- College Staff / Department Coordinator – verifies information and responds to queries referred by the chatbot.

Major use cases:

- Ask a Question – the user enters a question and receives a relevant response.
- View Suggested FAQs – the user selects a common question from the list.
- Ask a Follow-up Question – the user continues the conversation for clarification.
- Find Department Contact – the user receives the relevant office or department contact details.
- Manage FAQ – the administrator adds, updates, reviews, or removes FAQ information.
- Review Unanswered Questions – authorized staff review common unanswered queries and improve the information provided.

Textual use-case relationship: Students, prospective students, and parents interact with the chatbot to ask questions and view answers. Administrators sign in to manage FAQ content. College staff verify information and handle queries that require human assistance. A use-case diagram can represent these actors outside a system boundary labelled "AI-Based College FAQ and Student Support Chatbot," with the use cases listed above inside the boundary.

10. Use Case Descriptions

Use Case 1: Ask a Question
Actor: Student, prospective student, or parent.
Precondition: The chatbot interface is available.
Main flow: (1) The user opens the chatbot. (2) The user types a question. (3) The system processes the question. (4) The system searches approved information and prepares a response. (5) The answer is displayed to the user.
Alternative flow: If the question is unclear, the system requests clarification. If no reliable answer is found, the system provides the appropriate support contact.
Postcondition: The user receives an answer or a clear next step.

Use Case 2: Manage FAQ
Actor: Administrator.
Precondition: The administrator is authenticated and authorized.
Main flow: (1) The administrator opens FAQ management. (2) The administrator selects add, edit, or delete. (3) The administrator enters or updates the question and answer. (4) The system validates and saves the change. (5) The updated information becomes available to the chatbot.
Alternative flow: If required fields are missing or the data is invalid, the system displays an error and asks the administrator to correct it.
Postcondition: The approved FAQ information is stored and available for use.

11. Data Requirements

The system may store FAQ questions and answers, topic or category labels, department names, official contact details, update dates, administrator account details, and limited feedback or query statistics. Each FAQ record should have a unique identifier and a clear last-updated date. Only authorized users should be able to modify official content. Personal information should be minimized, access should be controlled, and records should be retained only as required by the college's policies.

12. External Interface Requirements

- User Interface: A chat window with a question-entry field, send button, answer area, suggested questions, and support-contact information.
- Administrator Interface: A protected page for signing in and managing FAQ content.
- Software Interface: The chatbot may connect to a database or approved knowledge base containing college information. If an AI language-processing service is used, communication should follow that service's documented interface.
- Hardware Interface: Users need a phone, tablet, or computer with a supported browser and network connection; the system requires suitable hosting infrastructure.
- Communication Interface: Data should be exchanged through secure network connections where supported.

13. Main System Features

- AI-supported question understanding and answer matching.
- Centralized college FAQ and student-support information.
- Suggested questions and follow-up conversation support.
- Department contact guidance for queries that cannot be resolved automatically.
- Administrator-controlled FAQ updates.
- Optional feedback and query analysis to identify missing information.

14. Assumptions and Constraints

Assumptions:

- The college will provide correct and approved information for the FAQ knowledge base.
- Users will have access to a compatible device and an internet connection.
- Authorized staff will review and update information when college details change.
- Users understand that the chatbot provides guidance and does not replace official decisions or human support.

Constraints:

- The chatbot's answer quality depends on the accuracy and completeness of the information provided.
- AI-based interpretation may not correctly understand every question, spelling variation, or language.
- Development is limited by available time, budget, staff, and technical resources.
- Use of third-party AI services, if any, depends on service availability, cost, and privacy requirements.
- The system must follow applicable college policies and data-protection requirements.

15. Requirement Completeness and Consistency Validation

The requirements should be reviewed with students, administrative staff, department coordinators, and the system administrator. Each requirement should be clear, necessary, testable, and consistent with the system scope. The following checks will help validate the SRS:

- Completeness: Confirm that common student questions, administrator tasks, and escalation paths are covered.
- Consistency: Ensure that the chatbot does not promise services that are outside its scope, such as changing marks or approving admissions.
- Clarity: Use specific wording so that each requirement has only one reasonable interpretation.
- Non-duplication: Remove repeated requirements that describe the same function.
- Testability: Define observable outcomes, such as whether an FAQ can be added or whether an unanswered question receives a support contact.
- Security and privacy: Verify that only authorized administrators can change official answers and that unnecessary personal data is not collected.
- Improvement: Test the chatbot with representative questions, review incorrect or unanswered results, and update the approved FAQ content.

16. Conclusion

The AI-Based College FAQ and Student Support Chatbot is intended to make college information easier to access and reduce the effort required to answer repeated questions. This requirement analysis identifies the system's stakeholders, scope, functional and non-functional requirements, actors, use cases, data, interfaces, assumptions, and constraints. Clear validation and regular updates to the approved knowledge base will help make the chatbot useful, reliable, secure, and easy to maintain.